

| Fundamental field                | Symmetries                                                                                                   | Decomposition into SO(3) irrep(s)                                   | Source                                         |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------|
| $\mathcal{Z}_\alpha$             | Symmetry[1, $\mathcal{Z}^{\bullet 1}$ , { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[]]          | $\mathcal{Z}^{\#1}_{1^-}_\alpha + \mathcal{Z}^{\#1}_{0^+} n_\alpha$ | $\mathcal{J}_{\mathcal{Z}_\alpha}$             |
| SO(3) irrep                      | Symmetries                                                                                                   | Expansion in terms of the fundamental field                         | Source SO(3) irrep                             |
| $\mathcal{Z}^{\#1}_{0^+}$        | Symmetry[0, $\mathcal{Z}^{\#1}_{0^+}$ , {}, StrongGenSet[{}], GenSet[]]                                      | $\mathcal{Z}^\alpha n_\alpha$                                       | $\mathcal{J}_{\mathcal{Z}^{\#1}_{0^+}}$        |
| $\mathcal{Z}^{\#1}_{1^-}_\alpha$ | Symmetry[1, $\mathcal{Z}^{\#1\bullet 1}_{1^-}$ , { $\bullet 1 \rightarrow -a$ }, StrongGenSet[{}], GenSet[]] | $\mathcal{Z}_\alpha - \mathcal{Z}^\beta n_\alpha n_\beta$           | $\mathcal{J}_{\mathcal{Z}^{\#1}_{1^-}_\alpha}$ |